

MICROCANONICAL EXCESS-PRESSURE SEMIGROUPS AND KINETIC THETA-GENERATORS

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ABSTRACT. The exact finite state is the probability law on the hard-sphere configuration augmented by its actual-contact ledger. A Yosida resolvent graph core and finite-generator correlation correctors give the Hamilton–Boltzmann nonlinear-generator limit. The transition action contains only dynamic entropy, and comparison is proved by bounded contact-ratio truncation followed by monotone removal of the truncation, yielding the unique kinetic semigroup and its microscopic limit.

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1. ROUND-NINE RECONSTRUCTION: AN EXACT LAW STATE, A RESOLVENT GRAPH CORE, AND ENTROPY-TRUNCATED COMPARISON

The exact finite system evolves on probability laws over the complete hard-sphere configuration and contact ledger. Its generator acts on observables on that law space. Static preparation is charged once at the initial boundary; every transition action contains only the dynamic contact entropy. The comparison theorem is proved first for bounded collision-control ratios and then by monotone entropy truncation, so large doubled-variable slopes never enter an uncontrolled exponential Hamiltonian.

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1.1. The exact augmented law state. Let $\mathbf{E}_\varepsilon$ be the graph-completed hard-sphere configuration space augmented by the cumulative marked contact measure up to the current time. Let

$$\mathcal{P}_\varepsilon = \mathcal{P}_W(\mathbf{E}_\varepsilon)$$

with the weighted weak topology. Liouville transport and deterministic update of the contact ledger define a linear Markov semigroup $\mathcal{U}_\varepsilon(t)$ on laws. On bounded cylindrical observables $\Phi(\rho)$ its generator is

$$\mathbb{A}_\varepsilon \Phi(\rho) = \lim_{t \downarrow 0} t^{-1} (\Phi(\mathcal{U}_\varepsilon(t)\rho) - \Phi(\rho)).$$

Theorem 1.1 (Exact augmented law semigroup). *The family $\mathcal{U}_\varepsilon(t)$ is a positive mass-preserving strongly continuous semigroup on $\mathcal{P}_\varepsilon$. The factorial correlation hierarchy and every ledger moment are compatible coordinates of a law, but are not themselves identified with the law state. The nonlinear log-Laplace semigroup*

$$\mathcal{V}_\varepsilon(t)F(\rho) = \mu_\varepsilon^{-1} \log \int e^{\mu_\varepsilon F(\eta)} d(\mathcal{U}_\varepsilon(t)\rho)(\eta)$$

satisfies the exact tower on the complete law state.

Proof. The hard-sphere flow is measurable and measure preserving on the full-measure regular graph-completed phase space. The contact ledger is an additive deterministic functional, so augmenting the state makes its update Markov. Push-forward gives the linear semigroup and continuity follows from the B2 Green-graph flow on weighted laws. The log-Laplace tower is the linear tower after exponentiation. Correlations determine a law only on the declared moment-determinate class and are used there solely as coordinates. $\square$

1.2. A genuine observable graph core. For $\lambda > 0$, let

$$R_{\lambda,\varepsilon} = \lambda \int_0^\infty e^{-\lambda t} \mathcal{U}_\varepsilon(t) dt$$

be the law-semigroup resolvent. Let $\mathcal{C}_\varepsilon$ be the algebra generated by

$$R_{\lambda,\varepsilon} \Phi, \quad \Phi(\rho) = \varphi(\rho(F_1), \dots, \rho(F_m)), \quad \varphi \in C_c^\infty,$$

where the F_j are smooth weighted configuration/contact cylinders.

Lemma 1.2 (Resolvent graph core). *The algebra $\mathcal{C}_\varepsilon$ is contained in $D(\mathbb{A}_\varepsilon)$, is stable under smooth bounded functional calculus, and is dense in the generator graph norm on every compact containment set. Its gradients and collision increments are bounded by constants determined by the chosen compact set and resolvent parameters.*

Proof. The resolvent identity gives $\mathbb{A}_\varepsilon R_{\lambda,\varepsilon} \Phi = \lambda(R_{\lambda,\varepsilon} \Phi - \Phi)$, so every generator image is explicit. The Markov chain rule for bounded smooth functions gives stability under functional calculus. Yosida approximations $\lambda R_{\lambda,\varepsilon} F$ converge to F and their generator images converge whenever F is in the

graph domain. Approximate the bounded cylinders uniformly on a compact containment set by smooth finite-coordinate cylinders; the Yosida identity upgrades this to graph-norm convergence. No analytic cutoff equal to one on an open set is used. $\square$

1.3. Finite-generator correlation correctors. Let F be a density/contact cylinder in the limiting core. Lift it to the augmented law state and expand its finite logarithmic generator in connected factorial coordinates. Write $D_j^\varepsilon(t)$ for the uncanceled j -label defect after the lower orders have been removed. Let $V_{j,\varepsilon}(t, s)$ be the backward observable propagator generated by the actual finite triangular BBGKY/ledger generator at order j .

Lemma 1.3 (Finite graph-domain corrector). *Define*

$$c_{j,\varepsilon}(t) = \int_t^T V_{j,\varepsilon}(t, s) D_j^\varepsilon(s) ds.$$

Then $c_{j,\varepsilon}(T) = 0$ and

$$\partial_t c_{j,\varepsilon} + A_{j,\varepsilon}(t) c_{j,\varepsilon} = -D_j^\varepsilon(t)$$

in the exact observable graph domain. On a slightly smaller hierarchy weight,

$$\sum_{j \geq 2} \mu_\varepsilon^{1-j} (\|c_{j,\varepsilon}\|_{mgr} + \|\mathbb{A}_\varepsilon c_{j,\varepsilon}\|_{mgr}) < \infty$$

uniformly on the kinetic horizon.

Proof. The B2 positive boundary-renewal hierarchy constructs $V_{j,\varepsilon}$ by forward Picard iteration in the remaining time and preserves the exact trace graph. Differentiating the Duhamel integral gives the terminal equation. A connected j -label defect has norm $j! C^j T^{j-1}$; the factorial law-coordinate weight and μ^{1-j} give a geometric majorant after one label-weight loss. The same bound applies after the finite generator because the Duhamel equation replaces its graph image by D_j and a time derivative. Thus the sum may be interchanged with the generator. $\square$

Theorem 1.4 (Microscopic nonlinear-generator convergence). *For every limiting core cylinder F there are corrected finite observables $F_\varepsilon \in D(\mathbb{A}_\varepsilon)$ with $F_\varepsilon \rightarrow F$ locally uniformly such that*

$$\frac{1}{\mu_\varepsilon} e^{-\mu_\varepsilon F_\varepsilon} \mathbb{A}_\varepsilon e^{\mu_\varepsilon F_\varepsilon} \longrightarrow \mathbb{H} \left(f, \frac{\delta F}{\delta f}, \psi_F \right)$$

locally uniformly on regular containment sets.

Proof. Sum the correctors from Lemma 1.3. Their finite graph images cancel every connected correlation defect. The remaining creation trees converge by the B2 source-uniform one-block theorem to the binary Hamilton–Boltzmann exponential. Cycles are $O(\varepsilon^\alpha)$. Normal graph convergence justifies exponentiation, differentiation, and interchange of the series with $\mathbb{A}_\varepsilon$. $\square$

1.4. Dynamic action and entropy truncation. For a balanced path on $[s, t]$ set

$$\mathcal{A}_{s,t}(f, \Gamma) = \int_s^t \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f dr.$$

It contains no $I_0(f_s)$. Static grand-canonical or microcanonical preparation is a boundary functional applied once at the initial time.

Theorem 1.5 (Additive kinetic transition semigroup). *For bounded uniformly continuous terminal data,*

$$(S_{s,t}\Phi)(f_s) = \sup_{(f,\Gamma): f(s)=f_s} \{\Phi(f_t) - \mathcal{A}_{s,t}(f, \Gamma)\}$$

is finite on weighted moment sublevels, attains its relaxed supremum, and satisfies

$$S_{r,t} = S_{r,s} S_{s,t}.$$

The full prepared value is $\sup_{f_0} \{S_{0,T}\Phi(f_0) - I_{0,a}(f_0)\}$.

Proof. B2 rate-sublevel compactness gives attainment. Splitting a balanced pair at s preserves the intermediate density/contact trace and adds the two dynamic entropy integrals; concatenation gives the reverse inequality. Since no new initial rate is charged on the second leg, there is no double counting. The last formula inserts preparation once. $\square$

For $M > 0$ restrict the contact ratio to $e^{-M} \leq q \leq e^M$. Its Hamiltonian is

$$\mathbb{H}_M(f, p) = \langle f, v \cdot \nabla p \rangle + \int \sup_{e^{-M} \leq q \leq e^M} \{q \Delta p - \ell(q)\} dA_f.$$

It is globally Lipschitz in the collision increment with constant e^M .

Lemma 1.6 (Comparison at fixed entropy truncation). *Bounded upper and lower viscosity solutions of*

$$\partial_t U + \mathbb{H}_M(f, DU) = 0$$

on a common moment sublevel satisfy comparison. The doubled-variable penalty can be chosen from finite weighted test coordinates and forces the diagonal while its cotangent remains in the bounded-increment core.

Proof. Use

$$\frac{1}{2\eta} \sum_{j=1}^J |\langle f - g, \phi_j \rangle|^2 + \delta(\Upsilon(f) + \Upsilon(g)).$$

At a positive maximum, the penalty itself is bounded; hence the coordinate distance is $O(\sqrt{\eta})$ and tends to zero. For fixed M the Hamiltonian continuity constant is finite, so the usual subtraction of the two viscosity inequalities closes after *arepsilon* $\downarrow 0$, then $J \rightarrow \infty$ and $\delta \downarrow 0$. Smooth saturation of each coordinate derivative keeps all collision increments in the fixed core without changing the penalty near its maximizing pair. $\square$

Theorem 1.7 (Untruncated comparison and nonlinear limit). *The semigroups S^M increase locally uniformly to S , and S is the unique bounded viscosity solution of the full Hamilton–Boltzmann equation in the entropy-coercive class. The exact microscopic log-Laplace semigroups converge locally uniformly to S . Microcanonical preparation is inserted only at the initial boundary.*

Proof. Perspective entropy makes paths with q outside $[e^{-M}, e^M]$ uniformly expensive on action sublevels. Truncating q and repairing the balance with the B2 conservative finite-cell correction gives $\mathcal{A}^M \downarrow \mathcal{A}$, hence $S^M \uparrow S$ locally uniformly. Comparison for every M and monotone stability give comparison for S . Theorem 1.4 gives upper and lower generator limits on the resolvent core, while B2 compact containment prevents escape. Half-relaxed limits are respectively sub- and supersolutions and are identified by comparison. B1 supplies the one-time prepared boundary functional. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

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